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PERSPECTIVES ON UTILITY THEORY

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Utility theory is a branch of decision analysis that is concerned with building models to explain and guide choice behavior under uncertainty in situations in which "long run" expected values are too simplistic. Beginning with Bernoulli in the 18th century, researchers have tried to analyze how people do and how people should make decisions when risk aversion is a factor. In this paper, we review some recent developments in the area.

Some exciting developments are taking place in the field of decision analysis. At the fall 1984 Dallas ORSA-TIMS Meeting, the 1983 Lanchester Prize Committee awarded Peter C. Fishburn an Honorable Mention for his research articles on transitive and nontransitive measurable utility theory. Fishburn has made many seminal contributions to the foundations of decision analysis over the past 20 years. His most recent papers reflect a trend of research that is being pursued by many individuals. This note offers some brief comments on these developments in decision analysis and provides a research bibliography of this activity.

For almost 40 years, decision analysts, economists, and others have relied on the axioms of von Neumann and Morgenstern (1947) to provide theoretical justification for using the principle of expected utility maximization in decision making under risk. In a recent overview of decision analysis published in *Operations Research*, Keeney (1982) discusses the importance of this approach and its many practical applications.

Although many quantitatively trained people find the von Neumann–Morgenstern axioms an acceptable basis for evaluating alternative decisions, empirical studies have commonly found that

(i) decision makers do not behave consistently with these axioms, and
(ii) when asked to reconcile their actual behavior with the axioms, many people (while accepting the appeal of the axioms) prefer to act in their original fashion.

The first finding is not serious. After all, many people are not good at multiplying three-digit numbers in their heads or at estimating the area of a tract of land. In many judgmental tasks, people do not automatically generate the correct answer. However, the second finding is more disturbing. Most people would readily defer to a calculator for solving the multiplication problem or to a surveyor for estimating acreage. Why do people not yield to the axioms?

Attempts to explain this last question can be grouped into the following categories.

(i) Decision makers who violate the axioms are being fooled by cognitive illusions, but unlike visual illusions, there is no way to demonstrate convincingly that the preference kind really are illusions.
(ii) Decision analysts have not considered all relevant dimensions of the risk-taking problem. Unless all features are accounted for, there is no way to reflect accurately a decision maker's preferences.
(iii) Decision analysts need to develop more flexible theories to accommodate actual decision behavior.

Fishburn's contributions are in the third category of

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research. His theories provide a weakening of the transitivity and independence axioms that are central to the von Neumann–Morgenstern theory. The following discussion gives a brief overview of this latter approach.

Expected Utility Theory

In an appendix to their book, *Theory of Games and Economic Behavior*, von Neumann and Morgenstern established a set of axioms for their theory of expected utility. These axioms have been refined by a number of researchers, so that the essential ideas of von Neumann and Morgenstern have been reduced to three basic axioms (e.g., Jensen 1967, Fishburn 1970). These axioms specify conditions on an individual's preference over pairs of risky prospects.

The first axiom requires that (1) for any pair of probability distributions p and q, either p is preferred to q ($p \succ q$), q is preferred to p ($q \succ p$), or the individual is indifferent between p and q ($p \sim q$) and (2) both the preference relation $\succ$ and the indifference relation $\sim$ are *transitive*. In other words, for any risky prospects p, q, and r, if $p \succ q$ and $q \succ r$, then $p \succ r$; likewise, if $p \sim q$ and $q \sim r$, then $p \sim r$. One can imagine violations of transitivity of preference when the prospects are evaluated on multiple criteria and violations of transitivity of indifference due to thresholds of discriminability. Although there have been extensions of expected utility that relax transitivity, this assumption has long been regarded as a tenet of rationality (e.g., see Tversky 1969, Machina 1981, and Fishburn 1981a).

The second axiom is called the *independence* axiom and is the source of much of the controversy about the von Neumann–Morgenstern theory and its refinements. Consider the following compound lottery. In the first stage, you obtain the rights to either the risky prospect p with probability λ or the risky prospect r with probability $1 - \lambda$. In the second stage, you determine which prospect you have. We denote this compound lottery by $\lambda p + (1 - \lambda)r$. Now compare this lottery with the compound lottery with q substituted for r, that is, $\lambda q + (1 - \lambda)r$. The independence axiom says that if p is preferred to q, then $\lambda p + (1 - \lambda)r$ is preferred to $\lambda q + (1 - \lambda)r$ for all prospects r and for all $0 < \lambda < 1$. Beginning with the classical paradox of Allais (1953), many behavioral researchers have documented systematic violations of the independence axiom across a range of subjects and conditions (e.g., see Allais and Hagen 1979, Hagen and Wenstop 1984, Hershey, Kunreuther, and Schoemaker 1982, Kahneman, Slovic, and Tversky 1982, MacCrimmon and Larsson 1979, Slovic and Tversky 1974, Stigum and Wenstop 1983, and others).

The third axiom is a mathematical assumption about continuity of preferences. These three axioms together imply the existence of a *utility function u* that has the following properties. First, the utility function preserves the order of preferences among risky prospects; that is, p is preferred to q if and only if the utility of p is greater than the utility of q: $p \succ q$ if and only if $u(p) > u(q)$. Second, the utility function is "linear in probabilities"; that is, $u(\lambda p + (1 - \lambda)q) = \lambda u(p) + (1 - \lambda)u(q)$. This linearity property is very important in decision analysis, because it permits one to evaluate compound lotteries easily by reducing them to an evaluation of their components. The familiar "folding-back" of a decision tree relies on this linearity property.

The main concern for operations researchers and decision analysts is what happens to the analytical convenience afforded by linearity of u if the independence axiom (or other axioms) are relaxed.

Skew-Symmetric Bilinear (SSB) Utility Theory

Fishburn (1982b, 1983a) provides a new theory of decision making under risk using axioms that are simultaneously weak enough to accommodate observed behavior and strong enough to have normative appeal. This so-called *skew-symmetric bilinear* (SSB) utility theory also answers many practical concerns for decision analysis.

Fishburn's theory derives its name from a skew-symmetric bilinear function ϕ, which is used to evaluate pairs of risky prospects as follows: $p \succ q$ if and only if $\phi(p, q) > 0$. (One can interpret $\phi(p, q)$ as a measure of the decision maker's intensity of preference for p over q, though many other interpretations are possible.) Skew-symmetry means that $\phi(p, q) = -\phi(q, p)$ and bilinearity means that $\phi(\lambda p + (1 - \lambda)q, r) = \lambda\phi(p, r) + (1 - \lambda)\phi(q, r)$. Bilinearity is a very important property, because it yields much of the analytical convenience of linearity for decision analysis applications.

The SSB theory that yields the functional ϕ rests on the following axioms. First, Fishburn uses a continuity axiom similar to the one used by von Neumann and Morgenstern. Second, the SSB theory relaxes the controversial independence axiom to give a *dominance* (or convexity) axiom, which says in part that if p is preferred to q and p is preferred or indifferent to r, then p is preferred to any compound lottery between q and r, $\lambda q + (1 - \lambda)r$ for all $0 < \lambda < 1$. Third, Fishburn uses a symmetry axiom that provides balance in indifference equations between compound lotteries. The axiom is somewhat involved to state, but it appears not to be controversial. Fishburn (1983a) gives a thoroughly readable account of these axioms and SSB utility theory.

The SSB utility theory does not require transitivity or the independence axiom. Therefore, it can account for a wide range of empirical observations. If in addition, we add the assumption that indifference be transitive, then ϕ can be decomposed in a *weighted linear utility representation*:

$$\phi(p, q) = u(p)w(q) - u(q)w(p).$$

Chew (1980, 1983), Chew and MacCrimmon (1979), and Nakamura (1984a, b) provide alternative axioms that yield this representation and give estimation procedures for the weight function w. Finally, if one were to add the independence axiom to the SSB axioms, then the von Neumann–Morgenstern utility function is produced from the relation $\phi(p, q) = u(p) - u(q)$, where the weights are all unity.

Related Research

Fishburn has applied SSB utility theory to a variety of issues in decision research, such as dominance analysis, preference reversals, multiattribute utility, and others. Related research on relaxing assumptions of von Neumann–Morgenstern utility theory includes Kahneman and Tversky's (1979) prospect theory, Chew (1980, 1983) and Chew and MacCrimmon's (1979) ratio model for weighted linear utility, Machina's (1982, 1983) generalized expected utility theory, Bell (1982, 1983) and Loomes and Sugden's (1982) regret theory, Nakamura's (1984a, b) extensions of nonlinear utility theory, Bell's disappointment model (1985a), and recent work by Luce and Narens (1985) on dual bilinear utility representations.

The elegance of the Fishburn axioms and the power of the representations make it likely that this research will become a benchmark for further studies of both prescriptive and descriptive decision making under risk. Many philosophical and practical questions remain to be answered about relaxations of the von Neumann–Morgenstern axioms. The following bibliography lists current research publications in this emerging area. The recent flurry of research is exciting. The field of decision analysis can look forward to continuing developments in the near future.

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